

Introduction to Financial Data

Every Investment Decision Starts with Data

Lecture 1 • Module 1: Understanding Financial Data

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Module 1

Understanding Financial Data

1.1 We Use Data Every Day

Everyday examples

-  Weather forecasts
-  Google Maps traffic
-  Sports statistics
-  Game rankings
-  YouTube views

Investor mindset

Investors also make decisions using data. They ask: **What changed?**

And: **Why did it change?**

Key idea: stock analysis is a data problem first, and a finance problem second.

1.2 What Is Financial Data?

Financial data describes how a company or market changes over time.

Example: a company like Tesla

Revenue

Profit

Employees

Stock Price

Which one changes every trading day?

👉 **Stock price**

Takeaway

Financial data helps us turn opinions into evidence.

1.3 What Is a Stock?

Simple definition

A stock means ownership in a company. A company is divided into shares, and people can buy or sell those shares.

Today's goal: understand the data behind a stock, not how to trade it.



Company



Shares



Price changes over time

1.4 Why Do Prices Change?

Possible reasons

-  Company news
-  Earnings reports
-  New products
-  World events
-  Economy changes

Class discussion

Why do you think Tesla's stock price changed so much during COVID-19?

We are not answering that fully yet. We are learning how to **collect the data** first.

Finding Financial Data

2.1 Where Can We Get Financial Data?

Need	Good source
Daily stock prices	Yahoo Finance
API access with free tier	Alpha Vantage, Finnhub
Fundamentals and filings	SEC EDGAR, StockAnalysis
Long-history research data	Shiller dataset, GuidingData
Simple downloads / CSV	Yahoo Finance, public datasets

Different websites specialize in different kinds of financial data. We choose the source based on the question we want to answer.

2.2 Yahoo Finance Walkthrough

Live demo steps

1. Search **TSLA**
2. Open the company page
3. View the chart
4. Open **Historical Data**
5. Choose a date range
6. Download CSV (Gold Membership required)

What you should notice

- stock price history is organized by date
- the data is already in table form
- this is the file we will load into Python later

2.3 Download with Python

Python can download the same data automatically.

Instead of clicking every time, we can use code to get a stock dataset and save it as a CSV file.

```
1  import yfinance as yf
2
3  tsla = yf.download(
4      "TSLA",
5      start="2020-01-01",
6      end="2025-01-01"
7  )
8
9  tsla.to_csv("tesla.csv")
```

In later lectures, we will learn what each line means.

Understanding a Stock Dataset

3.1 One Row = One Trading Day

Date	Open	High	Low	Close	Volume
2024-01-02	248.0	252.1	247.3	251.2	87,000,000

Important: this row represents one trading day, not one company.

Each column tells us something different about that day's trading activity.

3.2 Understanding the Columns

Date

Trading day

Open

First price

High

Highest price

Low

Lowest price

Close

Final price

Volume

Shares traded

Memory trick

Open = the first price

Close = the last price

High = the biggest price

Low = the smallest price

Volume = how active the trading was

3.3 Why Python?

Excel	Python
Hundreds of rows	Millions of rows
Manual work	Automated
Simple charts	Powerful visualization
Repeated tasks	Reusable programs

Python helps us work faster and think more clearly about the data.

3.4 Course Roadmap

1 **Find Financial Data** [START HERE](#)

2 Download CSV

3 Python (Pandas)

4 Visualization

5 Investment Metrics

6 **Portfolio Project**

Today's lesson is the starting point for everything that comes next.

Knowledge Check

Q1

Which website is commonly used for historical stock prices?

- A. YouTube
- B. Yahoo Finance
- C. Netflix

Q2

Which column shows the final trading price?

- A. Open
- B. Close
- C. Volume

Q3

One row usually represents:

- A. One company
- B. One trading day
- C. One year

Q4

Why use Alpha Vantage or Finnhub?

- A. Free API access
- B. Better videos
- C. Only cryptocurrency

Summary

Today you learned

-  What financial data is
-  Why stock prices change
-  Where financial data comes from
-  How to download stock data
-  What a stock dataset looks like

Big picture

We are building a workflow: **find data** → **organize data** → **visualize data** → **analyze data**.

Next Lecture

Downloading & Exploring Data

Next we will learn how to:

- Download CSV files
- Explore datasets in Excel
- Read data with Pandas
- Prepare data for analysis